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Salt bath compositions and methods for regenerating salt bath compositions

Abstract

Methods for regenerating a salt bath composition include heating the salt bath composition to an ion exchange temperature to form a molten salt bath. The methods may further include contacting at least a portion of an ion-exchangeable article that includes lithium oxide (Li.sub.2O) with the molten salt bath. Lithium cations may diffuse from the ion-exchangeable article and into the molten salt bath. Additionally, the methods may include adding a first phosphate salt to the molten salt bath. A lithium phosphate salt that includes at least a portion of the lithium cations may be formed and precipitate from the molten salt bath. Furthermore, the methods may include adding a multivalent salt that includes a multivalent metal cation to the molten salt bath. A second phosphate salt that includes the multivalent metal cation may be formed and precipitate from the molten salt bath.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a national stage entry of International Patent Application Serial No. PCT/US2020/050084 filed on Sep. 10, 2020, which claims the benefit of priority under 35 U.S.C. § 119 of U.S. Provisional Application Ser. No. 62/899,890 filed on Sep. 13, 2019 the content of which is relied upon and incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure generally relates to methods for regenerating lithium-enriched salt baths. More particularly, the disclosure relates to methods for using phosphate salts or a combination of phosphate salts and multivalent metal salts to remove undesirable ions from salt baths used in ion exchange processes for strengthening glass and glass-ceramic substrates while preventing or reducing the formation of surface defects.

BACKGROUND

(2) Tempered or strengthened glass is often used in consumer electronic devices, such as smart phones and tablets, due to its physical and chemical durability and toughness. In general, the durability of tempered glass and glass-ceramic substrates is increased by increasing the amount of compressive stress (CS) and depth of layer (DOL) of the glass or glass-ceramic substrate. To provide a larger CS and deepen the DOL, ion exchange processes may be used to strengthen glass or glass-ceramic articles. In ion exchange processes, a glass or glass-ceramic article containing at least one smaller alkali metal cation is immersed in a molten salt bath containing at least one larger alkali metal cation. The smaller alkali metal cations diffuse from the glass surface into the salt bath while larger alkali metal cations from the salt bath replace the smaller cations in the surface of the glass. This substitution of larger cations for smaller cations in the glass generates a layer of compressive stress layer in the surface of the glass, thus increasing the resistance of the glass to breakage.

(3) As the ion exchange proceeds, the salt concentration of the smaller alkali metal cations (i.e., the cations that diffuse from the glass into the salt) increases while the salt bath concentration of the larger alkali metal cations (i.e., the cations that migrate into the glass from the salt) decreases. This fluctuation in the ion concentration may cause unwanted constituents to form, which can age or “poison” the salt bath, or cause salt crystals to form and adhere to the surface of the glass. A poisoned salt bath will not produce a large CS and deep DOL in the glass substrate, as desired. Likewise, crystals on the surface of the glass can form defects—including depressions and protrusions. Dimpled, stippled glass, and glass that is not properly strengthened is not commercially desired and may be unsuitable for use in some industries

SUMMARY

(4) Embodiments herein address these needs by providing salt bath compositions and methods for regenerating salt bath compositions by selectively precipitating dissolved lithium cations from the salt bath to prevent and remedy salt bath poisoning while preventing or reducing surface defects.

(5) In embodiments, methods for regenerating a salt bath composition include heating the salt bath composition to an ion exchange temperature to form a molten salt bath. The methods may further include contacting at least a portion of an ion-exchangeable article that includes lithium oxide (Li.sub.2O) with the molten salt bath. Lithium cations may diffuse from the ion-exchangeable article and into the molten salt bath. Additionally, the methods may include adding a first phosphate salt to the molten salt bath. A lithium phosphate salt that includes at least a portion of the lithium cations may be formed and precipitate from the molten salt bath. Furthermore, the methods may include adding a multivalent salt that includes a multivalent metal cation to the molten salt bath. A second phosphate salt that includes the multivalent metal cation may be formed and precipitate from the molten salt bath.

(6) In embodiments, methods for regenerating a salt bath composition include heating the salt bath composition to an ion exchange temperature to form a molten salt bath. The methods may further include contacting at least a portion of an ion-exchangeable article that includes lithium oxide with the molten salt bath. Lithium cations may diffuse from the ion-exchangeable article and into the molten salt bath. Additionally, the methods may include adding a first phosphate salt having an average particle size greater than 100 micrometers (μm) to the molten salt bath. A lithium phosphate salt that includes at least a portion of the lithium cations may be formed and precipitate from the molten salt bath.

(7) In embodiments, salt bath compositions may include from 40 wt. % to 95 wt. % potassium nitrate based on the total weight of the salt bath composition, from 5 wt. % to 60 wt. % sodium nitrate based on the total weight of the salt bath composition, from 0.1 wt. % to 1 wt. % lithium nitrate based on the total weight of the salt bath composition, from 0.1 wt. % to 1 wt. % of a multivalent salt comprising a multivalent metal cation based on the total weight of the salt bath composition, and less than or equal to 0.01 wt. % of a first phosphate salt based on the total weight of the salt bath composition.

(8) In embodiments, salt bath compositions may include from 40 wt. % to 95 wt. % potassium nitrate based on the total weight of the salt bath composition, from 5 wt. % to 60 wt. % sodium nitrate based on the total weight of the salt bath composition, from 0.1 wt. % to 1 wt. % lithium nitrate based on the total weight of the salt bath composition, and from 0.1 wt. % to 1 wt. % of a phosphate salt. The average particle size of the phosphate salt may be greater than or equal to 100 μm .

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A schematically depicts a salt bath during an ion exchange process before lithium poisoning, according to embodiments disclosed and described herein;

(2) FIG. 1B schematically depicts a salt bath during an ion exchange process after lithium poisoning, according to embodiments disclosed and described herein;

(3) FIG. 2A schematically depicts a salt bath before regeneration of a lithium-poisoned salt bath, according to embodiments disclosed and described herein;

(4) FIG. 2B schematically depicts a salt bath after regeneration of a lithium-poisoned salt bath, according to embodiments disclosed and described herein;

(5) FIG. 3A schematically depicts a salt bath during regeneration, according to embodiments disclosed and described herein;

(6) FIG. 3B schematically depicts a salt bath during regeneration, according to embodiments disclosed and described herein;

(7) FIG. 4 schematically depicts an ion-exchangeable substrate during salt bath regeneration, according to embodiments disclosed and described herein;

(8) FIG. 5A schematically depicts an ion-exchangeable substrate in which a phosphate crystal has formed on the surface, according to embodiments disclosed and described herein;

(9) FIG. 5B schematically depicts a reaction occurring between a phosphate crystal and an ion-exchangeable substrate, according to embodiments disclosed and described herein;

(10) FIG. 6 schematically depicts an ion-exchangeable substrate comprising a surface defect, according to embodiments disclosed and described herein;

(11) FIG. 7 schematically depicts a power, a pellet, and a capsule of phosphate salt, according to embodiments disclosed and described herein;

(12) FIG. 8 is a contour plot of the compressive stress (CS) of ion-exchangeable substrates at different dosing concentrations of LiNO_3 and disodium phosphate (DSP), according to embodiments disclosed and described herein;

(13) FIG. 9 is a graph of the compressive stress of glass articles after each run in a salt bath containing 1 wt % trisodium phosphate and a bath containing 0.5% silicic acid, according to embodiments disclosed and described herein;

(14) FIG. 10 is a graph of the compressive stress of glass articles after each run in a salt bath containing 1 wt % trisodium phosphate following regeneration of the salt bath with the addition of 2 wt % trisodium phosphate, according to embodiments disclosed and described herein;

(15) FIG. 11 is a graph of the compressive stress of glass articles after each run in a salt bath containing 1 wt % trisodium phosphate compared to the compressive stress of glass articles after each run in a salt bath containing 2.5 wt % disodium phosphate and 0.5 wt % silicic acid, according to embodiments disclosed and described herein;

(16) FIG. 12 is a graph of the concentration of LiNO_3 present in two salt baths in which one bath contains trisodium phosphate and the other contains disodium phosphate, according to embodiments disclosed and described herein;

(17) FIG. 13 is a graph of the surface profile of an ion-exchangeable substrate having a surface defect, as measured using Zygo 3D imaging and surface metrology, according to embodiments disclosed and described herein;

(18) FIG. 14 is a second graph of the surface profile of a defect-free ion-exchangeable substrate, as measured using Zygo 3D imaging and surface metrology, according to embodiments shown and described herein;

(19) FIG. 15 is a graph of a high temperature x-ray diffraction analysis of a molten salt bath containing KNO_3 , NaNO_3 , and a small amount of lithium sodium phosphate, according to embodiments shown and described herein;

(20) FIG. 16 is a graph of the effective speed of the phosphate salt powder, pellet, and capsule, according to embodiments shown and described herein;

(21) FIG. 17 is a graph of the pH of a salt bath after adding various amounts of trisodium phosphate to the salt bath, according to embodiments shown and described herein;

(22) FIG. 18A is an optical image of a glass-ceramic article that has undergone an ion exchange process in a salt bath composition without a phosphate salt, according to embodiments shown and described herein;

(23) FIG. 18B is a scanning electron microscopy (SEM) image of the glass-ceramic article of FIG. 18A, according to embodiments shown and described herein;

(24) FIG. 18C is an optical image of a glass-ceramic article that has undergone an ion exchange process in a salt bath composition with a phosphate salt, according to embodiments shown and described herein;

(25) FIG. 18D is an SEM image of the glass-ceramic article of FIG. 18C, according to embodiments shown and described herein;

(26) FIG. 19A is an SEM image of a glass-ceramic article that has undergone an ion exchange process in a salt bath composition with both a phosphate salt and a multivalent metal salt, according to embodiments shown and described herein; and

(27) FIG. 19B is an SEM image of a glass-ceramic article that has undergone an ion exchange process in a salt bath composition with a phosphate salt, according to embodiments shown and described herein.

DETAILED DESCRIPTION

(28) Embodiments described herein are directed to methods for regenerating lithium-poisoned salt baths used in ion exchange processes to strengthen lithium-containing glass and glass-ceramic substrates while preventing or reducing the formation of surface defects.

(29) As used herein, the terms “ion exchange bath,” “salt bath,” and “molten salt bath,” are, unless otherwise specified, equivalent terms, and refer to the solution or medium used to effect the ion exchange process with a glass or glass-ceramic substrate, in which cations within the surface of a glass or glass-ceramic substrate are replaced or exchanged with cations that are present in the salt bath. It is understood that in embodiments the salt bath comprises at least one of KNO_3 and NaNO_3 , may be liquefied by heat or otherwise heated beyond a substantially solid phase.

(30) As used herein, the terms “substrate” and “article” are, unless otherwise specified, equivalent terms, referring to a glass or glass-ceramic material of any shape or form including, but not limited to, sheets, vials, and three dimensional glass articles.

(31) As used herein, the terms “cation” and “ion” are considered equivalent terms, unless otherwise specified. The terms “cation” and “ion” can also refer to one or more cations. While potassium and sodium cations and salts are used in embodiments, it is understood that all embodiments of the disclosure are not limited to these species. The scope of the present disclosure also includes other metal salts and ions, particularly cations and salts of the alkali metals, as well as those of other monovalent metals.

(32) As used herein, the terms “selectively” and “selective” are used to refer to the affinity for a product or reaction mechanism to be promoted, such that the particular product or reaction mechanism occurs over other potential products or reactions.

(33) As used herein, the term “diffusivity” refers to the molar flux of a species due to molecular diffusion or concentration gradient of the particular species.

(34) As used herein, the terms “granular” or “granulated” refers to a composition comprising distinguishable pieces or grains with discrete components.

(35) As used herein, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a” component includes aspects having two or more such components, unless the context clearly indicates otherwise.

(36) Specific embodiments will now be described with reference to the figures. The following description of the embodiments is illustrative in nature and is in no way intended to be limiting in its application or use. Furthermore, it should be understood that like reference numbers indicate corresponding or related parts in the various figures.

(37) FIGS. 1A and 1B schematically depict a salt bath **100** during an ion exchange process before and after lithium poisoning. The salt bath **100** contains larger alkali metal cations **120**, which will be exchanged with smaller alkali metal cations, depicted as lithium cations **130**, present in the ion-exchangeable substrate **105**. As the smaller alkali metal cations in the ion-exchangeable substrate **105** are exchanged with the larger alkali cations present in the salt bath **100**, a layer of compressive stress is generated in the surface of the ion-exchangeable substrate **105**. This compressive stress layer may inhibit both crack formation and crack propagation.

(38) In embodiments, the salt bath **100** may comprise at least one of KNO_3 and NaNO_3 . In embodiments, the salt bath **100** may comprise a combination of KNO_3 and NaNO_3 . The combination of KNO_3 and NaNO_3 may be selected based on the desired application. KNO_3 —when compared to NaNO_3 —comprises a larger alkali metal cation (i.e., K^{+} compared to Na^{+}) that will more readily exchange with larger alkali metal cations in the glass substrate, such as Na^{+} . Likewise, NaNO_3 —when compared to KNO_3 —comprises a smaller alkali metal cation (i.e., Na^{+} compared to K^{+}) that more readily exchange with

smaller alkali metal cations in the glass substrate. Accordingly, in embodiments, the concentrations of KNO_3 and NaNO_3 in the salt bath may be balanced based on the composition of the ion-exchangeable substrate **105** to provide an ion exchange process that increases both the CS and DOL in the ion-exchangeable substrate **105**. For example, the salt bath **100** may comprise from greater than or equal to 40 wt. % and less than or equal to 95 wt % KNO_3 , and from greater than or equal to 5 wt % and less than or equal to 60 wt % NaNO_3 . In embodiments, the salt bath **100** may comprise from greater than or equal to 45 wt % to less than or equal to 50 wt % KNO_3 , and from greater than or equal to 50 wt % to less than or equal to 55 wt % NaNO_3 . In embodiments, the salt bath **100** may comprise from greater than or equal to 75 wt % KNO_3 to less than or equal to 95 wt % KNO_3 and from greater than or equal to 5 wt % NaNO_3 to less than or equal to 25 wt % NaNO_3 . In embodiments, the salt bath **100** may comprise greater than or equal to 45 wt % KNO_3 and less than or equal to 67 wt % KNO_3 , and greater than or equal to 33 wt % NaNO_3 and less than or equal to 55 wt % NaNO_3 .

(39) In embodiments, the salt bath **100** may comprise lithium nitrate (LiNO_3). In embodiments, the salt bath **100** may comprise lithium nitrate in an amount of from 0.01 wt. % to 1 wt. %, from 0.01 wt. % to 0.8 wt. %, from 0.01 wt. % to 0.6 wt. %, from 0.01 wt. % to 0.3 wt. %, from 0.01 wt. % to 0.2 wt. %, from 0.01 wt. % to 0.1 wt. %, from 0.1 wt. % to 1 wt. %, from 0.1 wt. % to 0.8 wt. %, from 0.1 wt. % to 0.6 wt. %, from 0.1 wt. % to 0.3 wt. %, from 0.1 wt. % to 0.2 wt. %, from 0.2 wt. % to 1 wt. %, from 0.2 wt. % to 0.8 wt. %, from 0.2 wt. % to 0.6 wt. %, from 0.2 wt. % to 0.3 wt. %, from 0.3 wt. % to 1 wt. %, from 0.3 wt. % to 0.8 wt. %, from 0.3 wt. % to 0.6 wt. %, from 0.6 wt. % to 1 wt. %, from 0.6 wt. % to 0.8 wt. %, or from 0.8 wt. % to 1 wt. %, based on the total weight of the salt bath **100**. Without being bound by any particular theory, it is believed that when the concentration of lithium nitrate is too great the salt bath may become poisoned, as described herein, and adversely impact the ion exchange process. In contrast, when the concentration of lithium nitrate is too low, excess lithium ions may diffuse from substrates undergoing the ion-exchange process, which may result in the reduction of any crystallization in the ion-exchangeable substrate and an increase in sodium-rich regions.

(40) In embodiments, the salt bath **100** may comprise silicic acid. In embodiments, the salt bath **100** may comprise silicic acid in an amount of from 0.1 wt. % to 1 wt. %, from 0.1 wt. % to 0.8 wt. %, from 0.1 wt. % to 0.6 wt. %, from 0.1 wt. % to 0.4 wt. %, from 0.1 wt. % to 0.2 wt. %, from 0.2 wt. % to 1 wt. %, from 0.2 wt. % to 0.8 wt. %, from 0.2 wt. % to 0.6 wt. %, from 0.2 wt. % to 0.4 wt. %, from 0.6 wt. % to 1 wt. %, from 0.6 wt. % to 0.8 wt. %, or from 0.8 wt. % to 1 wt. %, based on the total weight of the salt bath **100**. Without being bound by any particular theory, it is believed that silicic acid may prevent the degradation of components of the salt bath, which adversely impacts the ion exchange process. For example, silicic acid may react with the decomposition products of potassium nitrate, such as potassium nitrite or potassium oxide, and prevent the formation of potassium hydroxide, which may etch and damage the surface of glass or glass-ceramic substrates.

(41) The ion exchange process may be promoted, in embodiments, by heating the salt bath **100** to an ion exchange temperature. When heated to an ion exchange temperature, the salt bath composition of salt bath **100** may be liquefied and form a molten salt bath suitable to promote the ion exchange process. If the temperature of the salt bath **100** is increased too much, it may be difficult to adequately control the ion exchange process and the DOL could increase too quickly without obtaining a good CS. Accordingly, the salt bath **100** may, in embodiments, be heated to a temperature of greater than or equal to 360° C. and less than or equal to 430° C. In embodiments, the salt bath **100** may be heated to a temperature of greater than or equal to 360° C. to less than or equal to 390° C., or of greater than or equal to 375° C. to less than or equal to 385° C., or of greater than or equal to 378° C. to less than or equal to 383° C., or of greater than or equal to 410° C. to less than or equal to 430° C. Alternatively, the salt bath **100** may be heated to a temperature of

greater than or equal to 400° C. to less than or equal to 430° C., or greater than or equal to 425° C. to less than or equal to 435° C. The salt bath **100** may be heated to a temperature of 380° C. or to a temperature of 420° C.

(42) In embodiments, such as those wherein the ion-exchangeable article comprises glass-ceramic, higher temperatures may be used to promote the ion exchange process (i.e., the ion exchange temperature) may be greater than 450° C. Accordingly, the salt bath **100** may, in embodiments, be heated to a temperature of from 450° C. to 550° C., from 450° C. to 525° C., from 450° C. to 500° C., from 450° C. to 475° C., from 475° C. to 550° C., from 475° C. to 525° C., from 475° C. to 500° C., from 500° C. to 550° C., from 500° C. to 525° C., or from 525° C. to 550° C. This increased temperature may promote an increase in the rate of diffusion of ion during ion exchange and decrease the time required to achieve the desired properties (i.e., CS and DOL) in ion-exchangeable substrates. However, in embodiments, the increased temperature may increase the solubility of phosphate salts, such as trisodium phosphate, and lithium phosphate salts, resulting in increased phosphate ions within the molten salt bath.

(43) In embodiments, before, during, or following the ion exchange process, the salt bath **100** may have a pH of from less than or equal to 10 and greater than or equal to 6, as measured by dissolving 5 wt % of the molten salt in aqueous solution and measuring the pH at room temperature (a temperature from 20° C. to 25° C., such as 21° C.). In embodiments, the salt bath **100** may be basic or alkaline, meaning that the salt bath may have a pH of greater than 7 at 25° C. In embodiments, the salt bath **100** may have a pH of from less than or equal to 10 to greater than or equal to 7.5 or from less than or equal to 10 and greater than or equal to 8, or from less than or equal to 10 and greater than or equal to 9, as measured by dissolving 5 wt % of the molten salt in aqueous solution and measuring the pH at room temperature.

(44) Although FIGS. 1A and 1B show the ion-exchangeable substrate **105** completely immersed in the salt bath **100**, it should be understood that in embodiments, only a portion of the ion-exchangeable substrate **105** is contacted with the salt bath **100**. The ion-exchangeable substrate **105** may be brought into contact with the molten salt through immersion in a salt bath **100**, or through spraying, dipping, or other similar means of contacting the ion-exchangeable substrate **105** with the salt bath **100**. In embodiments, the ion-exchangeable substrate **105** may be fully submerged in the salt bath **100** or only a portion of the ion-exchangeable substrate **105** may be submerged in the salt bath **100**. The ion-exchangeable substrate **105** may be brought into contact with the salt bath **100** multiple times, including but not limited to, dipping the ion-exchangeable substrate **105** into the salt bath **100**.

(45) At least a portion of the ion-exchangeable substrate **105** may be contacted with the salt bath **100** for a period of time of from 1 minute to 60 hours. At least a portion of the ion-exchangeable substrate **105** may be contacted with the salt bath **100** for a period of time of from 1 minute to 48 hours, or from 1 minute to 24 hours, or from 10 minutes to 24 hours, or from 10 minutes to 2 hours. In embodiments, the ion-exchangeable substrate **105** may be contacted with the salt bath **100** for from 5 hours to 40 hours, or from 5 hours to 20 hours, or for 8 hours to 24 hours. In embodiments, at least a portion of the ion-exchangeable substrate **105** may be contacted with the salt bath **100** for a period of time of from 10 minutes to 1 hour, or 10 minutes to 30 minutes, or from 1 hour to 3 hours.

(46) In embodiments, the ion-exchangeable substrate **105** is a glass or glass-ceramic substrate or article. The glass or glass-ceramic substrate may, in embodiments, comprise alkali aluminosilicate or alkali aluminoborosilicate glass. For example, in embodiments, the ion-exchangeable substrate **105** may be formed from a glass composition which includes greater than or equal to 50 mol % SiO₂ and less than or equal to 80 mol % SiO₂, greater than or equal to 0 mol % B₂O₃ and less than or equal to 5 mol % B₂O₃, greater than or equal to 5 mol % Al₂O₃ and less than or equal to 30 mol % Al₂O₃, greater than or equal to 2 mol % Li₂O and less than or equal to 25 mol % Li₂O, greater than or equal to 0 mol %

Na.sub.2O and less than or equal to 15 mol % Na.sub.2O, greater than or equal to 0 mol % MgO and less than or equal to 5 mol % MgO, greater than or equal to 0 mol % ZnO and less than or equal to 5 mol % ZnO, greater than or equal to 0 mol % SnO.sub.2 and less than or equal to 5 mol % SnO.sub.2, and greater than or equal to 0 mol % P.sub.2O.sub.5 and less than or equal to 10 mol % P.sub.2O.sub.5. Alternatively, the ion-exchangeable substrate may comprise greater than or equal to 60 mol % SiO.sub.2 and less than or equal to 75 mol % SiO.sub.2, greater than or equal to 0 mol % B.sub.2O.sub.3 and less than or equal to 3 mol % B.sub.2O.sub.3, greater than or equal to 10 mol % Al.sub.2O.sub.3 and less than or equal to 25 mol % Al.sub.2O.sub.3, greater than or equal to 2 mol % Li.sub.2O and less than or equal to 15 mol % Li.sub.2O, greater than or equal to 0 mol % Na.sub.2O and less than or equal to 12 mol % Na.sub.2O, greater than or equal to 0 mol % MgO and less than or equal to 5 mol % MgO, greater than or equal to 0 mol % ZnO and less than or equal to 5 mol % ZnO, greater than or equal to 0 mol % SnO.sub.2 and less than or equal to 1 mol % SnO.sub.2, and greater than or equal to 0 mol % P.sub.2O.sub.5 and less than or equal to 5 mol % P.sub.2O.sub.5. In embodiments, the ion-exchangeable substrate **105** may not comprise B.sub.2O.sub.3, P.sub.2O.sub.5, MgO, ZnO, SnO.sub.2, or combinations thereof. It should be understood that the aforementioned glass composition is one embodiment of a glass composition that may be used in the ion exchange process and that other lithium containing glass compositions for use with the methods described herein are contemplated and possible.

(47) As mentioned, in embodiments, the ion-exchangeable substrate **105** comprises lithium **130**. In embodiments, the ion-exchangeable substrate **105** may comprise greater than or equal to 2.0 mol % Li.sub.2O and less than or equal to 25 mol % Li.sub.2O. In other embodiments, the ion-exchangeable substrate **105** may comprise from greater than or equal to 2.0 mol % Li.sub.2O to less than or equal to 15 mol % Li.sub.2O, or from greater than or equal to 2.0 mol % Li.sub.2O to less than or equal to 10 mol % Li.sub.2O, or from greater than or equal to 2.5 mol % Li.sub.2O to less than or equal to 10 mol % Li.sub.2O. In embodiments, the ion-exchangeable substrate **105** may comprise from greater than or equal to 5.0 mol % Li.sub.2O to less than or equal to 15 mol % Li.sub.2O, or from greater than or equal to 5.0 mol % Li.sub.2O to less than or equal to 10 mol % Li.sub.2O, or from greater than or equal to 5.0 mol % Li.sub.2O to less than or equal to 8 mol % Li.sub.2O.

(48) The amount of lithium present in the ion-exchangeable substrate **105** may allow for the ion exchange process to occur at a faster rate when compared to the ion exchange processes of other ion-exchangeable substrates that do not contain lithium. In embodiments, lithium may more readily diffuse from the glass or glass-ceramic substrate and into the salt bath **100** than other alkali metal cations. In embodiments, the ion-exchangeable substrate **105** may diffuse lithium cations **130** from the substrate at a rate of greater than or equal to 1,000 square micrometers per hour ($\mu\text{m}^2/\text{hr}$). The ion-exchangeable substrate **105** may diffuse lithium cations **130** from the substrate at a rate of greater than or equal to 1,500 $\mu\text{m}^2/\text{hr}$, greater than or equal to 1,800 $\mu\text{m}^2/\text{hr}$, greater than or equal to 2,000 $\mu\text{m}^2/\text{hr}$, greater than or equal to 2,200 $\mu\text{m}^2/\text{hr}$, or greater than or equal to 2,500 $\mu\text{m}^2/\text{hr}$. In each of the above embodiments, the ion-exchangeable substrate **105** may diffuse lithium cations **130** at a rate of less than or equal to 8,000 $\mu\text{m}^2/\text{hr}$, such as less than or equal to 6,000 $\mu\text{m}^2/\text{hr}$ or less than or equal to 4,000 $\mu\text{m}^2/\text{hr}$. Although the diffusion rate has been described using lithium, it should be understood that the diffusion rate of any alkali metal cation may be adjusted.

(49) A fast diffusion rate of lithium cations **130** from the ion-exchangeable substrate **105** into the salt bath **100** may allow for a longer, and thus deeper, exchange of ions between the ion-exchangeable substrate **105** and the salt bath **100**. The diffusion rate of lithium cations **130** in the ion exchange process may govern the rate at which the ion-exchangeable substrate **105** is strengthened (by increasing CS and DOL) and generally decrease the time of the ion exchange process and/or the contact with the salt bath **100** to achieve a specific CS and/or DOL. Therefore, in embodiments, a high lithium cation **130** diffusion rate may be desired to decrease the time

required for the ion exchange process. However, in embodiments, a low lithium cation **130** diffusion rate may be desired to reduce or prevent the formation of surface defects on the ion-exchangeable substrate **105**, as the increased presence of ions in the salt bath **100** due to the fast diffusion rate may tend to propagate the formation of phosphate crystals, which may produce more numerous surface defects, more visible surface defects, or both.

(50) For example, in an embodiment, methods for reducing or preventing the formation of phosphate crystals, surface defects, or both, may include utilizing an ion-exchangeable substrate **105** with a low ion diffusion rate. The formation of phosphate crystals often occurs at the end of the ion exchange process after the concentration of lithium cations **130** present in the salt bath **100** has dissipated. Ion-exchangeable substrates **105** with low diffusion rates, such as a rate of less than or equal to 8,000 $\mu\text{m}\cdot\text{sup.2/hr}$ may allow for a slower, more shallow ion exchange, which may reduce or prevent phosphate crystal **247** formation. Therefore, in embodiments, the ion-exchangeable substrate **105** may have a diffusion rate of less than or equal to 8,000 $\mu\text{m}\cdot\text{sup.2/hr}$ to prevent the formation of phosphate crystals **247** and may be greater than or equal to 1,000 $\mu\text{m}\cdot\text{sup.2/hr}$ to allow the ion-exchangeable substrate **105** to be properly strengthened during the ion exchange process. The diffusion rate may be from 1,000 $\mu\text{m}\cdot\text{sup.2/hr}$ to 6,000 $\mu\text{m}\cdot\text{sup.2/hr}$, or from 2,000 $\mu\text{m}\cdot\text{sup.2/hr}$ to 6,000 $\mu\text{m}\cdot\text{sup.2/hr}$, or from 3,000 $\mu\text{m}\cdot\text{sup.2/hr}$ to 5,000 $\mu\text{m}\cdot\text{sup.2/hr}$. In embodiments, a diffusion rate of 8,000 $\mu\text{m}\cdot\text{sup.2/hr}$ may achieve an ion exchange depth of layer of 400 μm in 5 hours, which, in embodiments, may be the shortest time needed to achieve proper strengthening in an ion exchange reaction. In embodiments, a diffusion rate of 1,000 $\mu\text{m}\cdot\text{sup.2/hr}$ may achieve an ion exchange depth of layer of 400 μm in 40 hours, which may, in embodiments, be the longest time needed proper strengthening in an ion exchange reaction.

(51) The diffusion rate may be measured based on the equation listed below:

$$(52) \text{ DiffusionRate}(\mu\text{m}^2 / \text{hr}) = \text{DepthofLayer}(\mu\text{m}) \times \frac{\text{depthoflayer}(\mu\text{m})}{\text{timeelapsed}(\text{hr}) \times 3.92} \quad \text{EQUATION1}$$

(53) As mentioned above, lithium may more readily diffuse from the ion-exchangeable substrate **105** than other alkali metals. In embodiments, sodium may diffuse from the ion-exchangeable substrate **105** at a rate of from 100 $\mu\text{m}\cdot\text{sup.2/hr}$ to 800 $\mu\text{m}\cdot\text{sup.2/hr}$, or from 100 $\mu\text{m}\cdot\text{sup.2/hr}$ to 600 $\mu\text{m}\cdot\text{sup.2/hr}$, or from 200 $\mu\text{m}\cdot\text{sup.2/hr}$ to 500 $\mu\text{m}\cdot\text{sup.2/hr}$. Without being bound by any particular theory, the diffusion rate of the lithium cations **130**, sodium cations, or both, may be controlled to prevent poisoning and to reduce salt crystal formation in the salt bath **100**. In embodiments, too high of a concentration of lithium cations **130**, sodium cations, or both, may cause the fresh molten salt **101** to become poisoned or to form unwanted phosphate crystals, which may cause functional and/or aesthetic defects.

(54) Referring again to FIG. 1A, an ion-exchangeable substrate **105** immersed in a salt bath **100** is schematically depicted. In FIG. 1A, the salt bath **100** is a molten salt bath comprising KNO.sub.3 and NaNO.sub.3, such as described above. The salt bath **100** has fresh molten salt **101** and includes larger alkali metal cations **120**. The salt is considered to be “fresh” when the salt bath **100** has not been poisoned by smaller alkali metal cations, such as lithium cations **130**, which may diffuse from the ion-exchangeable substrate **105** and into the salt bath **100** to poison the fresh molten salt **101**. The larger alkali metal cations **120** may, in embodiments, comprise potassium, sodium, or combinations thereof, which may have disassociated from the KNO.sub.3 and NaNO.sub.3 present in the salt bath **100**.

(55) In embodiments, phosphate salts, such as trisodium phosphate, may be added to the salt bath **100** to precipitate out the excess lithium cations **130** before, during, or after the lithium cations **130** exchange out of the ion-exchangeable substrate **105**. The phosphate salt may be present in the salt bath **100** before the ion-exchangeable substrate **105** is contacted with the salt bath **100** or the phosphate salt may be added to the molten salt bath after at least one ion-exchangeable substrate **105** has contacted the molten salt bath **100**, or, in embodiments, both. In embodiments, the addition of a phosphate salt may reduce the concentration of lithium cations in the molten salt bath. For

example, in embodiments, the addition of a phosphate salt may reduce the concentration of lithium cations in the molten salt bath by at least 0.03 wt. %, based on the total weight of the salt bath composition and/or the molten salt bath. In embodiments, the addition of a phosphate salt may reduce the concentration of lithium cations in the molten salt bath by at least 0.05 wt. %, at least 0.07 wt. %, at least 0.09 wt. %, at least 0.1 wt. %, at least 0.13 wt. %, at least 0.15 wt. %, at least 0.17 wt. %, at least 0.19 wt. %, or at least 0.2 wt. %, based on the total weight of the salt bath composition and/or the molten salt bath.

(56) The phosphate salt may precipitate the lithium cations **130** to prevent the fresh molten salt **101** from becoming poisoned by having too high of a lithium concentration. However, after precipitation of the lithium cations **130**, the excess phosphate present in the salt bath **100** may form salt crystals, which may adhere to the surface of the ion-exchangeable substrate **105**. The phosphate crystals may interact with the ion-exchangeable substrate **105**, causing potassium cations, which may be preferably selected by the phosphate anions, to diffuse from the ion-exchangeable substrate **105**, while the sodium cations diffuse into the ion-exchangeable substrate. The increased concentration of sodium cations in the ion-exchangeable substrate **105** may create a depression in the ion-exchangeable substrate **105** caused by the volumetric difference between sodium cations and potassium cations.

(57) Without being bound by any particular theory, in embodiments, surface defects, such as depressions and protrusions, may be caused by crystals bonding to the ion-exchangeable substrate **105** following an ion exchange treatment. These surface defects may be visually observable with or without the use of a magnifying lens (such as a microscope) or a green light inspection process. The crystals may bond to the ion-exchangeable substrate **105** and remain on the ion-exchangeable substrate **105** following removal from the salt bath **100**. As the residual salt from the salt bath **100** cools on the ion-exchangeable substrate **105**, larger alkali metal cations **120** present in the crystals may exchange with smaller alkali metals such as the lithium cations **130** present in the ion-exchangeable substrate **105**. As the alkali metals exchange, localized stress may be formed in the ion-exchangeable substrate **105**, creating a dimensional difference in the surface of the ion-exchangeable substrate **105**. As the crystals are removed from the ion-exchangeable substrate **105** (such as by washing the ion-exchangeable substrate **105** with water), small depressions may remain in the surface of the ion-exchangeable substrate **105** caused by the exchange of cations from the crystal into the substrate. If the surface of the ion-exchangeable substrate **105** is thin, such as a thickness of less than or equal to 100 μm , when the crystals are removed from the ion-exchangeable substrate **105**, the voids left by the larger alkali metal cations **120** may be forced through the surface of the ion-exchangeable substrate **105** to form protrusions on the an opposite surface of the ion-exchangeable substrate **105**.

(58) In embodiments, portions of the phosphate salts, such as trisodium phosphate, which do not completely dissociate in the molten salt bath may interact with and/or attach to the surface of ion-exchangeable substrates, such as ion-exchangeable substrate **105**, within the molten salt bath. This may form a layer or regions of residue on the surface of the ion-exchangeable substrates, which at least partially block or hinder the diffusion of ions in and out of the ion-exchangeable substrate during the ion-exchange process. This residue may generate amorphous layers or regions on the ion-exchangeable substrate as well as physical stress that reduces the strength of the final product. Moreover, this residue may remain on the ion-exchangeable substrate after removal from the molten salt bath. Due to the basic nature of phosphate salts, the pH of any water films that form on the surface of the glass may increase. Such high pH water films may dissolve carbon dioxide present in the air and promote corrosion or cracking in sodium-rich regions of the ion-exchangeable substrate.

(59) FIG. 1B schematically depicts a poisoned molten salt **102**. A salt bath **100** is considered to have been “poisoned,” when lithium cations **130** have diffused from the ion-exchangeable substrate **105** into the salt bath **100** such that the poisoned molten salt **102** contains an increased

concentration of lithium cations **130**. Having lithium cations **130** present in the salt bath **100** in a concentration greater than 1 wt % lowers the CS and DOL of the ion-exchangeable substrate **105** when compared to an ion-exchangeable substrate **105** in a salt bath **100** having a concentration of lithium cations of less than or equal to 1 wt %, such as fresh molten salt **101**, as the increase of lithium cations **130** in the molten salt retards the ion exchange process. More particularly, a salt bath may become poisoned with lithium cations over time such that the ion exchange characteristics of the bath degrade over time and the strength characteristics, such as compressive stress, depth of layer, and ring on ring strength, of the ion-exchangeable substrates treated in the salt bath vary, creating inconsistent product attributes over a manufacturing run.

(60) Referring now to an embodiment shown in FIG. 2A, a salt bath **200** with poisoned molten salt **202** contains more than 1 wt % lithium cations **230** in the poisoned molten salt **202**. The salt bath **200** contains metal cations **220** similar to the salt bath **100** and larger alkali metal cations **120**, as described above with reference to FIGS. 1A and 1B. In FIG. 2A, a phosphate salt **240** is added to the salt bath **200** to regenerate the poisoned molten salt **202**. As mentioned, the phosphate salt **240** may be added before or after the ion-exchangeable substrate **205** is contacted with the salt bath **200**. The phosphate salt **240** comprises a cation and an anion that dissolves in the salt bath and disassociate to form PO_4^{3-} anions and cations (including, but not limited to, sodium or potassium ions). The dissolved PO_4^{3-} anions present in the salt bath **200** react with and selectively precipitate the dissolved lithium cations **230**, favoring a reaction with the lithium cations **230** over other potential reactions, such as a reaction with sodium cations or potassium cations in the salt bath **200**. The selective precipitation reaction produces insoluble Li_3PO_4 and Li_2NaPO_4 and LiNa_2PO_4 , and generates additional cations (including, but not limited, to sodium and potassium cations) in the salt bath **200**, which further aid in the ion exchange process between the ion-exchangeable substrate **205** and the salt bath **200**.

(61) To effectuate the selective precipitation of the lithium cations **230**, in embodiments, the phosphate salt **240** is added to the salt bath **200** so that the salt bath comprises greater than or equal to 0.10 wt % and less than or equal to 5.0 wt % of phosphate salt **240**. In embodiments, the salt bath **100** may comprise from greater than or equal to 0.50 wt % to less than or equal to 5.0 wt %, from greater than or equal to 0.50 wt % to less than or equal to 1.0 wt %, or from greater than or equal to 0.10 wt % to less than or equal to 5.0 wt % of phosphate salt **240**. In other embodiments, the salt bath **200** may comprise from greater than or equal to 0.10 wt % to less than or equal to 1.0 wt %, or from than or equal to 1.0 wt % to less than or equal to 5.0 wt % of phosphate salt **240**.

(62) In embodiments, the phosphate salt **240** may comprise Na_3PO_4 , K_3PO_4 , Na_2HPO_4 , K_2HPO_4 , $\text{Na}_5\text{P}_3\text{O}_{10}$, $\text{K}_5\text{P}_3\text{O}_{10}$, $\text{Na}_2\text{H}_2\text{P}_2\text{O}_7$, $\text{Na}_4\text{P}_2\text{O}_7$, $\text{K}_4\text{P}_2\text{O}_7$, $\text{Na}_3\text{P}_3\text{O}_9$, $\text{K}_3\text{P}_3\text{O}_9$, or combinations thereof. In embodiments, the phosphate salt **240** may comprise Na_3PO_4 , or may comprise K_3PO_4 , or may comprise a combination of Na_3PO_4 and K_3PO_4 . In embodiments, the phosphate salt **240** may comprise anhydrous trisodium phosphate (Na_3PO_4), which may contain 10% or less water and may have a chemical purity of at least 97% or greater. Trisodium phosphate may be commercially available as trisodium orthophosphate, anhydrous from Shifang Zhixin Chemical Co. Ltd., based in China, or Prayon Inc., based in the United States (Augusta, GA). In embodiments, the concentration of phosphate in the salt bath **200** may be greater than or equal to 50 parts per million (ppm) phosphate and less than or equal to 1,000 ppm phosphate. The concentration of phosphate in the salt bath **200** may be 500 ppm to 1,000 ppm phosphate or 250 ppm to 750 ppm phosphate.

(63) FIG. 2B shows an embodiment of a salt bath **200** with regenerated molten salt **211**. Particularly, the salt bath **200** has been regenerated by the addition of the phosphate salt **240** consequently precipitating lithium cations **230** out of the salt bath **200** in the form of insoluble

lithium phosphate (Li.sub.3PO.sub.4) **250**. Accordingly, the concentration of lithium cations **230** dissolved in the salt bath **200** is not greater than 1 wt % in the regenerated molten salt **211**. That is, the phosphate salt **240** has selectively precipitated the lithium cations **230** to form insoluble lithium phosphate **250**, reducing the lithium cations **230** dissolved in the salt bath **200** to less than 2 wt %. The dissociation of alkali metal cations (such as sodium or potassium) from the phosphate salt **240** in favor of the formation of lithium phosphate salt **250** also provides the salt bath **200** with more dissolved metal cations **220** that can later be exchanged into the ion-exchangeable substrate **205**. The lithium cations **230** present in the salt bath **200** may react with the nitrate NO.sub.3.sup.- anions present in the salt bath **200** to form lithium nitrate. In embodiments, the concentration of lithium nitrate in the salt bath **200** may be greater than or equal to 0 wt % and less than or equal to 2 wt %. In embodiments, the lithium nitrate concentration may be less than 1.5 wt % or less than 1 wt %, such as from greater than or equal to 0 wt % and less than or equal to 1.5 wt % or less than or equal to 1 wt % in the salt bath **200**. In embodiments, the lithium nitrate concentration in the salt bath **100** may be greater than or equal to 0 wt % to less than or equal to 0.5 wt %, greater than or equal to 0 wt % to less than or equal to 0.1 wt %, or greater than or equal to 0 wt % to less than or equal to 0.05 wt %. Alternatively, the lithium nitrate concentration in the salt bath **200** may be from greater than or equal to 0.5 wt % to less than or equal to 2 wt %, or from greater than or equal to 0.1 wt % to less than or equal to 1.5 wt %.

(64) In embodiments, the phosphate salt **240** may be added to the salt bath **200** such that the concentration of phosphate salt **240** is less than or equal to the concentration of the lithium cations **130** present in the salt bath **200**. Without being bound by any particular theory, excess phosphate salts **240** may, in embodiments, generate crystals on the surface of the ion-exchangeable substrate **105**. These crystals may propagate defects on the surface of the ion-exchangeable substrate **105**, which may not be desirable. By maintaining a concentration of phosphate salt **240** that is equal to or less than the concentration of the lithium cations **130**, the generation of surface defects may be reduced or prevented. This will be discussed further below with specific reference to FIGS. 4 to 6.

(65) In embodiments, the salt bath **200** may be used for at least 300 hours before the regenerated molten salt **211** needs to be changed. In other embodiments, the salt bath **200** may be used for at least 100 hours, 150 hours, 175 hours, 200 hours, 250 hours, or 400 hours before the regenerated molten salt **211** needs to be changed.

(66) As shown in the embodiments depicted by FIGS. 2A and 2B, the phosphate salt **240** may be added to “spike” the salt bath **200** without the ion-exchangeable substrate **205** present in the bath. For example, in embodiments, a poisoned salt bath may be spiked between ion exchange cycles. “Spiking” the salt bath **200** refers to adding phosphate salt to a poisoned salt bath. In embodiments, the phosphate salt **240** may be added in amounts from greater than or equal to 0.10 wt % to less than or equal to 10.0 wt %. In embodiments, the salt bath **200** may be spiked before the ion-exchangeable substrate **250** is contacted with the salt bath **200**, or after the ion-exchangeable substrate is contacted with the salt bath **200**, or both. For instance, in embodiments, a salt bath **200** comprising at least one of KNO.sub.3 and NaNO.sub.3 is heated and contacted with an ion-exchangeable substrate **205**. The compressive stress of the first ion-exchangeable substrate **205** may be measured through any techniques known in the industry. As the ion exchange process proceeds, the lithium cations **130** may begin to diffuse from the ion-exchangeable substrate **205**, poisoning the salt bath **200**. As the salt bath **200** becomes increasingly poisoned with the lithium cations **130**, the compressive stress of subsequent ion-exchangeable substrates **205** produced by contacting the substrate with the poisoned bath may begin to decrease as compared to the compressive stress of ion-exchangeable substrates produced by contacting the substrates with the non-poisoned bath. Once subsequently treated ion-exchangeable substrates **205** produced by contacting the substrate with the poisoned salt bath **200** have a compressive stress below a certain value relative to the ion-exchangeable substrates treated by contacting with a non-poisoned bath, phosphate salt **240** may be added to the bath. In embodiments, the ion-exchangeable substrates **205**

are removed from the salt bath **200**, the salt bath **200** may be spiked with phosphate salts **240** to mitigate the poisoning lithium cations **130** by selectively precipitating the lithium in the bath into sludge, which may then be removed from the bath.

(67) In embodiments, the at least one ion-exchangeable substrate **205** may be removed from the salt bath **200** when the compressive stress of the subsequent ion-exchangeable substrate **205** produced by contacting the substrate to the poisoned salt bath is measured to be from 40 to 70 MPa below the compressive stress of the first ion-exchangeable substrate **205** initially produced by contacting the substrate with the non-poisoned salt bath **200** before the lithium poisoning. In embodiments, the phosphate salt may be added to the bath when the compressive stress of subsequent ion-exchangeable substrates measures to be 50 to 70 MPa below the compressive stress of the ion-exchangeable substrates **205** initially produced by non-poisoned salt baths **200**, or from 50 to 60 MPa, or from 40 to 60 MPa below the compressive stress of the ion-exchangeable substrates **205** initially produced by non-poisoned salt baths **200**. Similarly, if the salt bath **200** initially contained phosphate salt **240** before the ion-exchangeable substrate **205** was contacted with the salt bath **200**, a second phosphate salt may be added to the salt bath when the compressive stress of the ion-exchangeable substrate **205** being treated in the salt bath **200** containing diffused lithium cations **130** is from 40 to 70 MPa less than a compressive stress of a substrate treated in a salt bath that does not contain diffused lithium cations. In embodiments, the phosphate salt may be added when the compressive stress is 50 to 70 MPa below the compressive stress of the ion-exchangeable substrates **205** initially produced by non-poisoned salt baths **200**, or from 50 to 60 MPa, or from 40 to 60 MPa below the compressive stress of the ion-exchangeable substrates **205** initially produced by non-poisoned salt baths **200**. In such embodiments, the second phosphate salt may have the same composition or a different composition as the phosphate salt initially contained in the salt bath.

(68) In embodiments, the phosphate salt **240** may be added to the salt bath **200** when the CS of the ion-exchangeable substrate **205** treated in the poisoned salt bath **200** containing diffused lithium cations **130** (hereinafter referred to as “second CS”) is from 10 MPa to 70 MPa below the CS of a similar ion-exchanged substrate in the non-poisoned salt bath **200** before the diffusion of lithium cations **130** (hereinafter referred to as “second CS”). Lithium cations **130** may be selectively precipitated from the salt bath **200** when the ion-exchangeable substrate **205** has a compressive stress of from 10 MPa to 70 MPa less than the compressive stress of a similar ion-exchangeable substrate **205** treated in a salt bath **200** that does not contain diffused lithium cations. To precipitate the lithium cations **130**, phosphate salt **240** may be added to the salt bath **200** when the average second CS is approximately 50 to 60 MPa, or 30 to 40 MPa, or 45 to 55 MPa below the first CS of the ion-exchangeable substrate **205**. In other embodiments, the phosphate salt **240** may be added when the average second CS is approximately 20 to 30 MPa, or 30 to 50 MPa, or 40 to 45 MPa, or 45 to 50 MPa below the first CS.

(69) In embodiments, phosphate salt **240**, such as TSP, may be added to the salt bath **200** until the second CS is within 10 to 20 MPa of the first CS of the non-poisoned ion-exchangeable substrate **205**. Without being bound by any particular theory, by not adding enough TSP to bring the CS back to the original level in the non-poisoned ion-exchangeable substrate **205** may aid in reducing or preventing the formation of phosphate crystals **247**. In embodiments, phosphate salts **240** may be added until the second CS is within 15 to 25 MPa of the original CS of the ion-exchangeable substrate **205**, or until the CS is within 5 to 10 MPa, or within 15 to 20 MPa, or within 10 to 15 MPa, or within 15 to 25 MPa of the first CS of the non-poisoned ion-exchangeable substrate **205**. In embodiments, phosphate salt **240** may be added to the salt bath **200** to maintain a CS of from 10 to 50 MPa below the first CS of the ion-exchangeable substrate **205**. In embodiments, phosphate salt **240** may be added to the salt bath **100** to maintain a second CS of from 10 to 40 MPa, or 20 to 50 MPa, or 20 to 40 MPa, or 15 to 45 MPa, or 5 to 40 MPa below the first CS of the non-poisoned ion-exchangeable substrate **205**.

(70) The amount of phosphate salts **240** added to the salt bath **100** may depend on a variety of factors, including the ion-exchangeable substrate **105** and the size of the salt bath **100**. In embodiments, adding 0.5 wt % TSP may equate to a CS recovery of approximately 30 MPa. Therefore, in embodiments, for a salt bath **100** that holds approximately 1400 kg of salt, 0.5 wt % of 1400 kg would be equivalent to 7 kg of TSP needed to recover the CS of the ion-exchangeable substrate **105** by an increase in CS of about 30 MPa.

(71) In embodiments, before, during, and/or after the addition of phosphate salt **240**, the salt bath **100** may be inspected for visual quality. If the salt bath **200** is still cloudy, such as from the addition of phosphate salt **240**, it may stain the ion-exchangeable substrate **205** and may, in embodiments, create a haze on the ion-exchangeable substrate **205**. Hazy or stained ion-exchangeable substrates **205** may not be suitable for use in particular industrial applications. In embodiments, the salt bath **200** may be inspected for visual quality before the ion-exchanged substrate **205** is contacted with the salt bath **200** to prevent haze and/or staining.

(72) In embodiments, the phosphate salt **240** may be added to the salt bath **200** in a substantially solid phase before the ion-exchangeable substrate **205** is added to the salt bath **200**. For instance, the phosphate salt **240** may be added to the salt bath **200** as a powder, a pellet, a capsule, or in granular form. In embodiments, the phosphate salt **240** may be added to the salt bath **200** as a powder to promptly regenerate the salt bath **200** due to the high surface-to-volume ratio of powders. In other embodiments, the phosphate salt **240** may be added to the salt bath **200** in a capsule form to promptly sink to the bottom of the salt bath **200**. The advantages and disadvantages of adding the phosphate salt **240** in a powdered, pellet, and capsule form will be discussed further below with reference to FIGS. 7 and 16.

(73) After the phosphate salt **240** is added to the salt bath **200**, the phosphate salt **240** may settle in the salt bath **200**. Some phosphate salt **240** may substantially dissolve to a liquid phase in the salt bath **200**, which may allow for stratification to occur between the lithium cations **230** and the liquefied phosphate salt **240**. In embodiments, the salt bath **200** may comprise both liquid phosphate salt **240** which may disassociate into PO_4^{3-} anions and cations, and solid phosphate salt **240**, which may be allowed to settle into the bottom of the salt bath **200** before the ion-exchangeable substrate **205** is contacted with the salt bath **200**, so as to prevent defects from forming in the ion-exchangeable substrate **205**. In embodiments, the disassociated phosphate salt **240** may stratify and selectively precipitate the lithium cations **230**. In embodiments, as lithium phosphate **250** is selectively precipitated from the salt bath **200**, the phosphate salt **240** may transition from a substantially solid phase at the bottom of the salt bath **200** to a substantially liquid phase, disassociating into PO_4^{3-} anions and cations to allow for further selective precipitation with the lithium cations **230**. Thus, it should be understood that the liquid phase of the phosphate salt **240** may react with the lithium cations **230** in a liquid phase to form insoluble (i.e., solid) lithium phosphate **250** which may precipitate from the salt bath **200**.

(74) Embodiments further comprise forming insoluble lithium phosphate **250** from the selective precipitation reaction, as discussed above, and removing the insoluble lithium phosphate **250** from the salt bath **200**. In embodiments, the lithium phosphates **250** may settle to the bottom of the salt bath **200**, where the lithium phosphate **250** may be cleared or removed with a filter, or a sieve or strainer, or removed through other means. Alternatively, the insoluble lithium phosphate **250** may remain in the bottom of the salt bath **200**.

(75) Referring now to FIGS. 3A and 3B, a salt bath **300** is schematically depicted during regeneration. FIG. 3A depicts a salt bath **300** with fresh molten salt **301**, metal cations **320**, and that comprises phosphate salt **340** in addition to the at least one of KNO_3 and NaNO_3 . The ion-exchangeable substrate **305** contains lithium cations **330**, which will diffuse out of the ion-exchangeable substrate **305** and into the salt bath **300**. As shown in FIG. 3B, due to the phosphate salts **340** present in the fresh molten salt **301**, as the lithium cations **330** diffuse from the ion-exchangeable substrate **305**, the lithium cations **330** will be selectively precipitated by the

phosphate salt **340** to form insoluble lithium phosphate **350**, which sinks to the bottom of the salt bath **300**. The phosphate salt **340** selectively precipitates the lithium cations **330** to a concentration of less than 2 wt % in the salt bath **300**, and the fresh molten salt **301** of FIG. 3A becomes regenerated molten salt **311** as the lithium cation **330** concentration is greater than or equal to 0 wt % and less than or equal to 2 wt % lithium cations **330**.

(76) As shown in FIGS. 3A and 3B, in embodiments the salt bath **300** does not need to be spiked, as discussed above. In embodiments, the phosphate salt **340** may be added to the salt bath **300** to form lithium phosphate **350** as the lithium cation **330** concentration increases during the ion exchange process. This allows for the ion exchange process to occur at a faster rate. Furthermore, it allows for the regenerated molten salt **311** to be changed less frequently.

(77) Referring now to FIG. 4, a schematic view of an ion-exchangeable substrate **105** in a salt bath **100** is depicted. As phosphate salt **240** is added to the salt bath **100**, the lithium cations **130** present in the salt bath **100** are selectively precipitated to form lithium phosphates **250**. In embodiments, the lithium phosphates **250** may be a mixed lithium phosphate **242**, such as lithium sodium phosphate, $\text{Li.sub.2NaPO.sub.4}$ or $\text{LiNa.sub.2PO.sub.4}$, or combinations thereof. Lithium cations **130** preferably bond with the phosphate salt **240** over other alkali metal cations present in the salt bath (such as the metal cations **220**). However, as the lithium cation **130** concentration diminishes in the bath, the phosphate salt **240** may begin to react with other alkali metal cations present in the salt bath, such as medium alkali metal cations **140**. The medium alkali metal cations **140** may, in embodiments, be larger than the lithium cations **130** but smaller than the larger alkali metal cations **120**. The medium alkali metal cations **140** may be Na.sup.+ , K.sup.+ , or other alkali metals such as Rb.sup.+ , Cs.sup.+ , or Fr.sup.+ .

(78) Without being bound by any particular theory, as the lithium cations **130** are precipitated out of the salt bath **100**, the remaining phosphate salt **240** present in the salt bath **100** may be attracted to the medium alkali metal cations **140** to form phosphate alkali salts **246**. The phosphate alkali salts **246** may comprise trisodium phosphate (“TSP,” Na.sub.3PO.sub.4) or tripotassium phosphate (K.sub.3PO.sub.4). The phosphate alkali salts **246** may be the same or may be a different composition than the originally added phosphate salts **240**. In embodiments, the phosphate alkali salts **246** may not precipitate to the bottom of the salt bath **100** as lithium phosphates **250** and phosphate alkali salts **246**. Rather, in embodiments, the phosphate alkali salts **246** may instead remain in the salt bath **100**.

(79) Referring now to FIG. 5A, in embodiments, the phosphate alkali salts **246** form phosphate crystals **247**, which may attach to the surface of the ion-exchangeable substrate **105**, as depicted by arrow **249**. The phosphate crystals **247** may accumulate to form “sludge” in the salt bath **100**. The sludge may be a liquid slurry comprising phosphate crystals **247** and other bath contaminants. The salt may be periodically analyzed to determine the amount and content of the sludge in the salt bath **100**. In embodiments, the sludge may be physically or chemically removed from the salt bath **100**. For instance, the sludge may be physically skimmed out of the bath, or the sludge may be filtered from the bath. The formation of sludge may, in embodiments, be proportional to the amount of phosphate salt **240** added to the salt bath **200**. For instance, in embodiments, 1 kg of TSP may produce about 0.8 kg of sludge. This weight discrepancy may be due to the exchange of sodium in the TSP for lithium cations **130** which may precipitate to form $\text{Li.sub.2NaPO.sub.4}$, $\text{LiNa.sub.2PO.sub.4}$, Li.sub.3PO.sub.4 , or combinations thereof. In embodiments, the rate at which phosphate salt **240** is added to the salt bath **200** may be proportional to the rate at which sludge may be removed from the salt bath **200**.

(80) Upon removal of the sludge, it may be necessary to add additional salt to the salt bath **100** to maintain the proper salt ratio in the bath. In embodiments, salt may be added to restore the ratio of the salts in the salt bath **100** to the original salt ratio. For instance, if the salt bath originally contained 49 mol % NaNO.sub.3 and 51 mol % KNO.sub.3 , and upon removal of sludge some salt was lost from the bath such that the ratio of salt was now 43 mol % NaNO.sub.3 and 57 mol %

KNO₃, NaNO₃ may be added to the salt bath to raise the content of NaNO₃ by 6 mol %. This may prevent the salt concentration from impacting the ion exchange process of the ion-exchangeable substrate **205**, which may cause unwanted changes in the properties of the substrate.

(81) Still referring to FIG. 5A, the medium alkali metal cations **140** (such as sodium cations) present in the phosphate alkali salts **246** may begin to undergo an ion-exchange process with the ion-exchangeable substrate **105** with the larger alkali metal cations **120** (such as potassium cations) and any remaining lithium cations **130** present in the ion-exchangeable substrate **105**. The medium alkali metal cations **140** in the phosphate crystals **247** may exchange into the surface of the substrate, while the larger alkali metal cations **120** and any remaining lithium cations **130** may exchange out of the substrate and into the phosphate crystals **247**. While the lithium cations **130** may form insoluble lithium phosphate, Li₃PO₄, or insoluble lithium sodium phosphate, Li₂NaPO₄, or insoluble lithium disodium phosphate, LiNa₂PO₄, and precipitate, the large alkali metal cations **120** may form phosphate salts, (such as tripotassium phosphate, K₃PO₄) which remain as a phosphate crystal **247**.

(82) In FIG. 5B, once the ion-exchangeable substrate **105** is removed from the salt bath **100**, the crystals are removed from the ion-exchangeable substrate **105**, depicted by arrow **280**. The crystals may be removed **280** by washing the ion-exchangeable substrate **105** with water, such as deionized water, or by any known suitable method. The phosphate crystals **247** may now contain the larger alkali metal cations **120** and lithium cations **130** bonded with the phosphate salts **240** to produce, for instance, lithium sodium phosphate Li₂NaPO₄ or lithium disodium phosphate LiNa₂PO₄ or potassium sodium phosphate K₂NaPO₄ in the phosphate crystals **247**. As the phosphate crystals **247** are removed, the ion-exchangeable substrate **105** may be left with a surface depression where the phosphate crystals **247** previously resided on the ion-exchangeable substrate **105**.

(83) As discussed previously, as the alkali metals exchange between the ion-exchangeable substrate **105** and the phosphate crystals **247**, the difference in volumetric size between the larger alkali metal cations **120**, the medium alkali metal cations **140**, and the lithium cations **130** may generate localized stress in the ion-exchangeable substrate **105**. The localized stress may create a dimensional difference in the surface of the ion-exchangeable substrate **105** and may form a surface defect by leaving depression from the volumetric void previously occupied by the larger sized cation. Ideally, ion-exchange process generates a uniform compressive layer of stress in the ion-exchangeable substrate **105** without regions of localized, highly concentrated areas of stress. Highly concentrated localized regions of stress may generate defects in the ion-exchangeable substrate **105**, including depressions and protrusions in the surface of the ion-exchangeable substrate **105**.

(84) As shown in FIG. 6, the removal of the phosphate crystals **247** may reveal surface defects **440** on the surface of the ion-exchangeable substrate **105**, to form a defective ion-exchanged substrate **108**, which may not be desirable. Without being bound by any particular theory, in the surface of the ion-exchangeable substrate **105**, the void from the larger alkali metals (such as the larger alkali metal cations **120**) may not be filled completely by the smaller alkali metals (such as the medium alkali metal cations **140**), and as the phosphate crystals **247** are removed from the ion-exchangeable substrate **105** small depressions in the surface of the defective ion-exchanged substrate **108** may be formed. In embodiments, a depression generated in one surface of a substrate may produce a protrusion on an opposite surface of the substrate. This may particularly occur in thin substrates, such as substrates having a thickness of less than or equal to 100 μm, or less than or equal to 150 μm, or less than or equal to 75 μm, or less than or equal to 50 μm.

(85) In embodiments, the surface defect **440** may be a depression or a protrusion having a height or depth of greater than or equal to at least 1.2 nm and a width or length of greater than or equal to at least 0.005 mm. The surface defect **440** may have a height or depth of from 1 nm to 200 nm, or

from 1 nm to 100 nm, or from 1 nm to 10 nm. In embodiments, the surface defect **440** may have a height or depth of from 10 nm to 50 nm, or from 100 nm to 50 nm, or from 100 nm to 200 nm. The surface defect **440** may have a width or length of from 0.1 μm to 5 μm , or from 0.1 μm to 50 μm , or from 0.1 μm to 0.5 mm. In embodiments, the surface defect **440** may have a width or length of from 0.1 to 0.5 mm, or from 0.05 to 0.1 mm, or from 0.05 to 0.5 mm, or from 0.001 to 0.5 mm. In embodiments, the surface defects **440** may be visible to the unassisted eye. In other embodiments, the surface defects **440** may only be visible when placed under a green light or using a microscope or another magnifying lens.

(86) Various strategies may be employed to prevent or reduce the formation of the surface defects **440**, the phosphate crystals **247**, or both. In embodiments, as previously mentioned, it may be desirable to limit the concentration of phosphate salts **240** added to the salt bath **100** such that the concentration of lithium cations **130** is greater than or equal to the concentration of the phosphate salt **240** present in the salt bath **100**. As the phosphate salt **240** will preferably attract lithium cations **130**, which may precipitate out of the bath as solid lithium phosphates **250**, by limiting the concentration of phosphate salt **240** present in the bath to less than or the same as the concentration of the lithium cations **130**, the phosphate crystals **247** may not form, thereby preventing the surface defects **440**. While a minority of phosphate salt **240** may still react with the larger alkali metal cations **120** and the medium alkali metal cations **140** to form phosphate crystals **247**, the reduced amount of phosphate crystals may be easily washed from the ion-exchangeable substrate **105** without causing visible, or in embodiments, any surface defects **440**.

(87) One or more embodiments for reducing or preventing phosphate crystals **247**, surface defects **440**, or both, includes the addition of a multivalent metal salt to the salt bath composition and/or the molten salt bath to remove excess phosphate ions that may be present in the molten salt bath. The multivalent metal salt may be added before or after an ion-exchangeable article is contacted with the molten salt bath. The multivalent metal salt, may dissolve in the molten salt bath and the ions of the multivalent metal salt may disassociate to form anions and multivalent metal cations (including, but not limited to, magnesium, calcium, aluminum, strontium, barium, iron, and combinations thereof). The dissolved multivalent metal cations present in the molten salt bath may react with and precipitate the dissolved PO_4^{3-} anions from the molten salt bath. The selective precipitation reaction produces insoluble phosphate salts (including, but not limited to, magnesium phosphates, calcium phosphates, aluminum phosphates, and combinations thereof) from the molten salt bath. In embodiments, the addition of a multivalent metal salt may reduce the concentration of PO_4^{3-} anions in the molten salt bath. For example, in embodiments, the addition of a multivalent metal salt may reduce the concentration of PO_4^{3-} anions in the molten salt bath to less than or equal to 0.01 wt. %, less than or equal to 0.008 wt. %, less than or equal to 0.006 wt. %, less than or equal to 0.004 wt. %, or less than or equal to 0.002 wt. %, based on the total weight of the salt bath composition and/or the molten salt bath.

(88) To effectuate the precipitation of the potassium cations, in embodiments, the multivalent metal salt is added to the salt bath compositions and/or the molten salt bath such that the salt bath composition includes the multivalent metal salt in an amount, as determined by the following equation:

$$[\text{MMS}] = 1.5 \times ([\text{PS}] - ([\text{LiNO}_3]_{\text{sub. initial}} - [\text{LiNO}_3]_{\text{sub. Final}}) \times 1.2) \quad \text{EQUATION 2}$$

wherein “[MMS]” is the concentration of the multivalent metal salt, “[PS]” is the concentration of the phosphate salt in the salt bath composition and/or molten salt bath, “[LiNO₃]_{sub. Initial}” is the initial concentration of lithium nitrate in the salt bath composition and/or the molten salt bath as measured before the addition of the phosphate salt, and “[LiNO₃]_{sub. Final}” is the final concentration of lithium nitrate in the salt bath composition and/or the molten salt bath as measured after the addition of the phosphate salt. For example, if 0.2 wt. % trisodium phosphate is added to a molten salt bath and reduces the concentration of lithium nitrate from 0.2 wt. % to 0.09 wt. %, Equation 2 dictates that approximately 0.1 wt. % of a multivalent metal salt should be added to the

molten salt bath in order to effectuate the precipitation of the potassium cations that remain in the molten salt bath after regeneration. In embodiments, the multivalent metal salt may be added to the salt bath composition and/or the molten salt bath in an amount from 0.01 wt. % to 0.5 wt. %, from 0.01 wt. % to 0.4 wt. %, from 0.01 wt. % to 0.3 wt. %, from 0.01 wt. % to 0.2 wt. %, from 0.01 wt. % to 0.1 wt. %, from 0.01 wt. % to 0.05 wt. %, from 0.05 wt. % to 0.5 wt. %, from 0.05 wt. % to 0.4 wt. %, from 0.05 wt. % to 0.3 wt. %, from 0.05 wt. % to 0.2 wt. %, from 0.05 wt. % to 0.1 wt. %, 0.1 wt. % to 0.5 wt. %, from 0.1 wt. % to 0.4 wt. %, from 0.1 wt. % to 0.3 wt. %, from 0.1 wt. % to 0.2 wt. %, 0.2 wt. % to 0.5 wt. %, from 0.2 wt. % to 0.4 wt. %, from 0.2 wt. % to 0.3 wt. %, 0.3 wt. % to 0.5 wt. %, from 0.3 wt. % to 0.4 wt. %, or from 0.4 wt. % to 0.5 wt. %, based on the total weight of the salt bath composition and/or the molten salt bath.

(89) As previously mentioned, the multivalent metal salt may include magnesium cations, calcium cations, aluminum cations, and combinations thereof. For examples, the multivalent metal salts may include magnesium nitrate, calcium nitrate, aluminum nitrate, and combinations thereof. In embodiments, the multivalent metal salt may be anhydrous. As used herein, the term “anhydrous” may refer to a multivalent metal salt that includes less than or equal to 10% water and has a chemical purity greater than or equal to 97%. Without being bound by any particular theory, the use of an anhydrous multivalent metal salt may avoid any unwanted reactions between the water and the molten salt bath during the addition of the multivalent metal salt. In embodiments, such as those where the multivalent metal salt is not anhydrous, the multivalent metal salts may be dried at, for example, 300° C. for 2 hours to evaporate excess water prior to addition to the salt bath composition and/or molten salt bath.

(90) Embodiments for reducing or preventing phosphate crystals **247**, surface defects **440**, or both, include adding a phosphate salt having an average particle size greater than or equal to 100 μm, as measured by Brunauer-Emmett-Teller (BET) nitrogen absorption. For example, the phosphate salt may have an average particle size greater than or equal to 150 μm, greater than or equal to 200 μm, greater than or equal to 250 μm, greater than or equal to 300 μm, greater than or equal to 350 μm, greater than or equal to 400 μm, greater than or equal to 450 μm, or greater than or equal to 500 μm. Without being bound by any particular theory, it is believed that phosphate salts having a larger average particle size may stratify to the bottom of the ion exchange salt bath (away from the ion-exchangeable substrates), which reduces interaction between the phosphate salts and the surfaces of the ion-exchangeable substrates. This prevents the formation of residue on the surface of the ion-exchangeable substrates that may adversely impact the ion exchange process and/or the strength of the ion-exchangeable substrates after the ion exchange process. Moreover, without being bound by any particular theory, it is believed that a larger average particle size may reduce the solubility of the phosphate salt in the molten salt bath and, in turn, reduce the amount of excess phosphate ions in the molten salt bath, which may form phosphate crystals on the surface of the ion-exchangeable substrates.

(91) Another embodiment for reducing or preventing phosphate crystals **247**, surface defects **440**, or both, includes reducing the amount of time in which the ion-exchangeable substrate **105** is in contact with the phosphate salt **240**. For instance, the formation of phosphate crystals **247**, surface defects **440**, or both, may be reduced or removed by limiting the amount of time in which the ion-exchangeable substrate **105** is present in the salt bath **100**. This may prevent the formation of the phosphate crystals **247** in the salt bath **100**, or may reduce the concentration of the phosphate crystals **247** in the salt bath **100**, both of which may reduce the likelihood of the phosphate crystals **247** attaching on the surface of the ion-exchangeable substrate **105**.

(92) Likewise, the formation of phosphate crystals **247**, surface defects **440**, or both, may be reduced or removed by quickly cooling the ion-exchangeable substrate **105** upon removal from the salt bath **100**. In embodiments, the ion-exchangeable substrate **105** may be cooled from the temperature of the salt bath **100** to a temperature of less than or equal to 100° C. in less than 5 minutes. In embodiments, the ion-exchangeable substrate **105** may be cooled to a temperature of

less than or equal to 100° C. in less than 3 minutes, or less than 2 minutes, or less than 1 minute, or less than 1 minute and 30 seconds, or less than 30 seconds, or less than 10 seconds. Without intent to be bound by any particular theory, if the ion-exchangeable substrate **105** is cooled down to a temperature of less than or equal to 100° C. in greater than 5 minutes, the phosphate crystals **247** may be able to coalesce or otherwise grow larger in size, which may propagate more surface defects **440**, more visible surface defects **440**, or both. In embodiments, the ion-exchangeable substrate **105** may be cooled from a temperature of greater than or equal to 380° C. to a temperature of less than or equal to 100° C. in from 30 seconds to 3 minutes, or from 30 seconds to 1 minute and 30 seconds. In embodiments, the ion-exchangeable substrate **105** may be cooled to a temperature of less than or equal to 100° C. in from 1 minute to 3 minutes, or from 1 minute and 30 seconds to 3 minutes, or from 45 seconds to 2 minutes or from 2 minutes to 3 minutes.

(93) Additionally, phosphate crystals **247** have a propensity to crystallize at low temperatures due to their low solubility. In embodiments, the formation of phosphate crystals **247**, surface defects **440**, or both may be reduced or prevented by utilizing a salt bath **100** with a temperature of at least 380° C., such as at least 400° C., 420° C., or 450° C. The temperature of the salt bath **100** may be dependent on the composition of the ion-exchangeable substrate **105**.

(94) As previously mentioned, the method of introducing the phosphate salt **240** to the salt bath **100** may affect the reactions between the lithium cations **130** and the phosphate salt **240**, thus impacting the formation of the phosphate crystals **247** or lack thereof. As shown in FIG. 7, the phosphate salt **240** may be added to the salt bath **100** as a powder **810**, as a pellet **830**, or as a capsule **850**. As used here, a “powder” refers to fine dry particles produced by the grinding, crushing, or disintegration of a solid substance. As used herein, a “pellet” refers to a compressed shaped mass of a powder. As used herein, a “capsule” refers to a substance, such as a powder, that is substantially enclosed and enveloped by a sheath or membrane.

(95) Conventionally, powdered substances may be preferred due to the high surface-to-volume ratio that allows a powder **810** to react quickly and efficiently with the other reactants, such as the lithium cations **130**. However, adding powder **810** to the salt bath **100** may result in a cloudy bath in which the powder has stratified due to the small individual particles, as powder may have a tendency to create dust during dispensing that must be managed during application. In embodiments, a salt bath **100** that is cloudy may not be suitable for an ion exchange process, as it may impart various defects in the ion-exchangeable substrate **105**. The salt bath **100** may remain cloudy for a long period of time, such as 6 hours, 8 hours, 10 hours, one day or three days or more, prolonging the process and increasing the time and cost required to process the ion-exchangeable substrate **105**. Conventionally, for these reasons, a pellet **830** may be preferably used, as a pellet **830** may sink to the bottom of the salt bath **100** without stratifying and producing cloudiness in the salt bath **100**. However, a pellet **830** may not react as quickly with the lithium cations **130** due to the decreased surface area of the pellet when compared to the powder **810**.

(96) Therefore, in embodiments, a capsule **850** may be used to introduce the phosphate salt **240** to the salt bath **100**. In embodiments, the capsule **850** may be comprised of powdered phosphate salts **240** encapsulated with a salt mixture. In embodiments, the salt mixture may comprise NaNO_3 , KNO_3 , or both. Upon introduction to the salt bath **100**, the capsule **850** may sink to the bottom of the salt bath **100** where the encapsulated salt may dissolve, releasing the inner powdered phosphate salt **240** to quickly react with the lithium cations **130** without stratifying and causing a cloudy salt bath **100**.

(97) To prepare a capsule **850**, in embodiments, a phosphate salt may be blended with an encapsulating salt. The blended salts may be subjected to heat at a temperature above the melting temperature of the encapsulating salt such that the encapsulating salt begins to fuse. The heated salt mixture may be cooled to produce an encapsulated phosphate salt, referred to as capsule **850**.

(98) Similarly, in embodiments, granular TSP may be used. Granular TSP may not produce as much dust during dispensing and may prevent the bath from otherwise clouding. Granulated TSP

may have a coarse texture that quickly settles in the bath, and in embodiments, granular TSP may more readily dissolve than the encapsulated powder based on the size and surface area of the coarse particles. However, granular TSP may absorb moisture faster than encapsulated TSP during storage. (99) In embodiments, the phosphate salt may be a sodium phosphate, such as trisodium phosphate or disodium phosphate. The encapsulating salt may be a nitrate salt, such as potassium nitrate or sodium nitrate. The encapsulating salt may be used in fine crystal, pellet, or granular form. In embodiments, the encapsulating salt may have a lower melting temperature than the phosphate salt, such that upon heating the encapsulating salt melts and fuses around the substantially non-melted phosphate salt to create a capsule **850**. In embodiments, the melted salt may be cooled to room temperature (21° C.) or may be cooled at a temperature sufficient to form the capsule **850**. For instance, the melted salt may be cooled to any temperature below the glass transition temperature (T.sub.g) of the encapsulating salt. As shown below with reference to FIG. **17**, in embodiments, the capsule **850** may be able to react with lithium cations **130** at substantially the same rate as the powder **810**.

(100) The following examples illustrate one or more embodiments of the present disclosure as previously discussed above. The description of the embodiments is illustrative in nature and is in no way intended to be limiting its application or use.

EXAMPLES

Example 1—Lithium Poisoned Molten Salt of Sodium Nitrate

(101) A molten salt of sodium nitrate (NaNO₃) containing 0.32 wt % of lithium nitrate (LiNO₃) was characterized by inductively coupled plasma optical emission spectrometry (ICP-OES). Into this salt, 1 wt % of sodium phosphate (Na₃PO₄) was added to precipitate the lithium cations at 380° C. The treated molten salt was sampled and characterized by ICP-OES. Results showed that lithium cations were successfully removed from molten salt (liquid phase). The concentration of LiNO₃ was below the detection limit of ICP-OES (<0.005 wt %).

Example 2—Lithium Poisoned Molten Salt of Sodium Nitrate and Potassium Nitrate

(102) The molten salt of 49 wt % of sodium nitrate (NaNO₃) and 51 wt % of potassium nitrate (KNO₃) contained 0.02 wt % of lithium nitrate (LiNO₃) characterized by ICP-OES. Into this salt, 1 wt % of trisodium phosphate (TSP, Na₃PO₄) was added to precipitate the lithium cations at 380° C. The treated molten salt was sampled and characterized by ICP-OES. Results showed that lithium cations were successfully removed from molten salt (liquid phase). The concentration of LiNO₃ was below the detection limit of ICP-OES (<0.005 wt %).

Example 3—Lithium Poisoned Molten Salt of Sodium Nitrate and Potassium Nitrate

(103) Mixtures of 49 wt % of sodium nitrate (NaNO₃) and 51 wt % of potassium nitrate (KNO₃) containing 0.5-1.0 wt % of lithium nitrate (LiNO₃) and 0-2.5 wt % of disodium phosphate (DSP, Na₂HPO₄) were melted at 380° C. for 12 hours. Then, glass substrate (0.8 mm thickness, lithium-containing glass) were strengthened (ion-exchanged) in these baths for 3 hr. and 45 min. at 380° C. The strengthened glasses were cleaned and the compressive stress (CS) was measured by FSM (see Table 1, below). The lithium concentrations (in terms of LiNO₃) of molten salt samples were analyzed by ICP-OES (see Table 1, below).

(104) Results showed that lithium cations can be effectively stratified by the addition of phosphate (in this case, DSP). ICP data indicates lithium cation concentrations can be significantly reduced within 12 hours at 380° C. Using these baths for ion exchange, the CS was partially or completely recovered with the addition of DSP. Overdosing DSP helps a faster stratification process (see FIG. **8**).

(105) FIG. **8** is a contour plot of the compressive stress (CS) of ion exchanged glass substrates at different dosing concentrations of LiNO₃ and disodium phosphate (DSP), as discussed in detail in Example 3. FIG. **8** depicts five ion exchanged glass substrates with differing surface compressive stresses due to different ion exchange conditions (i.e., the concentration of DSP and lithium nitrate salt in the bath). The five ion exchange glasses have compressive stresses of 445.0

megapascals (MPa), 455.0 MPa, 465.0 MPa, 475.0 MPa, and 485.0 MPa. The contour plot shows the dosing concentrations of LiNO_3 in wt % versus the concentration of disodium phosphate (DSP) in wt %. As can be seen from FIG. 8, the compressive stress of the ion-exchanged substrates generally decreases as LiNO_3 content increases.

(106) TABLE-US-00001 TABLE 1 Stratification of Lithium Cations by DSP Average CS of Soluble the Added Added LiNO_3 in strengthened LiNO_3 DSP bath after 12 glass Conditions (wt %) (wt %) hours (wt %) (MPa) 1 0 0.5 0 495 2 0 0 0 490 3 0.5 0 0.43 450 4 0.5 0.5 0.3 465 5 0.5 0.5 0.22 465 6 1 0.5 0.72 440 7 0.5 2.5 0.05 487 8 1 2.5 0.13 470

(107) For all conditions the concentration of KNO_3 and NaNO_3 were 51 wt %, and 49 wt %, respectively.

Example 4—Trisodium Phosphate Bath Regeneration

(108) To determine the effectiveness of trisodium phosphate (TSP, Na_3PO_4) in precipitating lithium to regenerate a salt bath, 1 wt % TSP was pre-loaded in a 3 kg molten salt bath comprising 49 wt % NaNO_3 and 51 wt % KNO_3 . The salt bath was maintained at a temperature of 380° C. and 25 samples of Example Glass 1, a glass composition according to the embodiments described herein (composition described below in Table 2), were added to the bath. The glass pieces were 50 mm×50 mm×0.8 mm sheets of glass, inserted into the salt bath for 3 hr. and 45 min. The ion exchange process was repeated for 15 runs. For purposes of comparison, a second salt bath comprising 16 kg of 49 wt % NaNO_3 and 51 wt % KNO_3 with 0.5 wt % silicic acid (Si(OH)_4) was melted to replicate a standard commercial ion exchange bath. The second salt bath was maintained at a temperature of 380° C. and 120 samples of Example Glass 1 were added to the bath and ion exchanged for 3 hr. and 45 min. The ion exchange process was repeated for 6 runs. The compressive stress (CS) of the glass samples was determined after each run as a measure of the degree of poisoning of the salt bath. That is, it was expected that the compressive stress of the samples would decrease for each consecutive run due to the increasing concentration of lithium in the bath as a result of ion exchange.

(109) FIG. 9 shows the compressive stress of the glass samples after each run of Example 4 for the bath containing 1 wt % TSP and the bath containing 0.5 wt % silicic acid. As shown in FIG. 9, the bath containing 1 wt % TSP generally yielded higher compressive stresses than the bath containing 0.5 wt % silicic acid. The data also indicates that the decrease in compressive stress in the bath containing 0.5 wt % silicic acid was more pronounced (i.e., the CS curve had a steeper slope) as the number of runs increased than the bath containing 1 wt % TSP. While not wishing to be bound by theory, it is believed that this trend is the result of lithium poisoning in the bath containing 0.5 wt % silicic acid while the bath containing 1 wt % TSP at least partially mitigated the effects of lithium poisoning.

(110) In addition, to determine the effectiveness of regenerating the salt bath containing 1 wt % TSP, an additional 2 wt % TSP was added to the salt bath containing TSP after run 15. This “spike” of TSP regenerated the salt bath by precipitating out the poisoning lithium cations. As shown in FIG. 10, after run 15, when the additional 2 wt % TSP was added to the salt bath, the compressive stress of the glass samples increased from 460 MPa to 500 MPa. This demonstrates that, after regeneration, the salt bath is able to produce glass pieces with similar compressive stresses as those in the original, fresh salt bath that has not been poisoned (i.e., as shown in FIG. 9).

(111) Regeneration of the bath by the addition of TSP produced glass samples with compressive stresses over 440 MPa after 30 runs in the same salt bath. In prior ion exchange processes (i.e., without regeneration), the salt bath would need to be changed after only 7-8 runs or batches. By increasing the time and the amount of batches of glass that may be ion exchanged before the molten salt bath must be cooled, cleaned, replaced and reheated, production efficiencies are improved and production costs are decreased.

(112) TABLE-US-00002 TABLE 2 Composition of Example Glass 1 Example Mol % Glass 1 Oxide SiO_2 63.46 Al_2O_3 15.71 B_2O_3 0 Li_2O 6.37 Na_2O 10.69

Example 5—Trisodium Phosphate Pre-Loading

(113) To determine the amount of TSP that could be pre-loaded into the salt bath to prevent lithium poisoning, three salt bath tanks were artificially poisoned with LiNO₃, as shown in Table 3 below. In the first salt bath, labeled “Tank 1,” 1.56 wt % of LiNO₃ was used to artificially poison a 3 kg bath comprising 49% NaNO₃ and 51% KNO₃. The bath was maintained at a temperature of 380° C. 2.5 wt % TSP was added to the poisoned bath. Once the constituents had settled in the bath, 25 pieces of glass according to Example Glass 1 were added. The glass pieces were ion exchanged for 3 hr. and 45 min. The compressive stress in the glass was measured after ion exchange and determined to be 466.7 MPa, an acceptable value for commercial production.

(114) Similarly, in the second salt bath, labeled “Tank 2,” 6.25 wt % of LiNO₃ was used to artificially poison a 3 kg bath comprising 49% NaNO₃ and 51% KNO₃. The bath was maintained at a temperature of 380° C. 10 wt % TSP was added to the poisoned bath. Once the constituents had settled in the bath, 25 pieces of glass according to Example Glass 1 were added to the bath and ion exchanged for 3 hr. The compressive stress of the glass was measured after ion exchange and determined to be 464.7 MPa with no cloudiness on the glass surface to indicate impurities.

(115) The data in Table 3 demonstrates that large amounts of lithium poisoning may be overcome by pre-loading up to 10 wt % TSP in the salt bath to prevent lithium poisoning. More specifically, the data demonstrates that the lithium poisoning can be overcome by maintaining the ratio of LiNO₃ to TSP as the concentration of LiNO₃ in the bath increases without adversely affecting the compressive stress imparted to the glass. The data also confirmed that by pre-loading up to 10 wt % TSP into the salt bath, the life of the salt bath could be extended by at least 70 runs without the need for regeneration.

(116) TABLE-US-00003 TABLE 3 Compressive Stress using 2.5 wt % and 10 wt % TSP

Number of Tank/Salt	TSP LiNO ₃ / LiNO ₃	TSP CS	Experiment	Pieces	Vol. (kg)	(g) TSP (wt %)/ (ppm)	(MPa)
Artificially poisoned	25	3	75	0.62	0.28	130	466.7
Tank 1, with 2.5% TSP, 1.56% LiNO ₃							
Artificially poisoned	25	3	300	0.62	0.23	<50	464.7
Tank 2, with 10% TSP, 6.25% LiNO ₃							

Example 6—Disodium Phosphate Bath Regeneration

(117) To determine the effectiveness of disodium phosphate (DSP, Na₂HPO₄) in precipitating lithium to regenerate a salt bath, 2.5 wt % DSP was pre-loaded in a 3 kg molten salt bath comprising 49 wt % NaNO₃ and 51 wt % KNO₃. The salt bath was maintained at a temperature of 380° C. and 25 samples of Example Glass 1, a glass composition according to the embodiments described herein (composition described above), were added to the bath. The glass pieces were 50 mm×50 mm×0.8 mm sheets of glass, inserted into the salt bath for 3 hours and 45 minutes.

(118) FIG. 11 shows a comparison between Example 4 (TSP Regeneration), Example 5 (DSP Regeneration), and silicic acid only (no phosphate addition). As shown in FIG. 11, DSP is slightly less effective than TSP in salt bath regeneration, but is more effective than silicic acid alone.

(119) TSP exhibited superior regeneration capability when compared to DSP over longer periods of time, as shown FIG. 12. Particularly, FIG. 12 shows the results of two salt baths of 49% NaNO₃ and 51% KNO₃ at 380° C. Both salt baths contained 0.495 wt % LiNO₃ pre-loaded to artificially poison the salt baths. To the poisoned salt bath of Tank 1, 0.37 wt % DSP was added. To the poisoned salt bath of Tank 2, 0.43 wt % TSP was added (equivalent to a 1.1 molar ratio of DSP and TSP). The wt % of LiNO₃ in the bath was tested over a period of hours. While DSP and TSP show similar results after 5-10 hours, after 24 hours, TSP shows a decreased level of LiNO₃ when compared to DSP. Further, after a period of 5 days (120 hours), the tank containing TSP has a lower concentration of LiNO₃ when compared to the tank containing DSP.

Example 7—Surface Defects

(120) FIG. 13 depicts a characterization of the surface defects present on a glass substrate following an ion exchange process in which the salt bath was regenerated with trisodium phosphate salt (TSP). FIG. 13 is a graph of the Zygo measurements, a 3-dimensional (3D) imaging and surface metrology tool which depicts the surface profile of the glass sheet. As the ion exchange process occurred, the TSP began to precipitate out the lithium cations to form lithium phosphate salts. As the lithium concentration diminished, the excess TSP began to react with the potassium present in the glass substrate, causing an influx of sodium cations into the glass substrate, while phosphate crystals comprising K_3PO_4 were formed. Upon removing the glass article from the bath and cooling the glass article, water was used to try to wash off the phosphate crystals. As the large phosphate crystals were removed from the glass substrate, surface defects were noticed on the glass substrate. As shown in FIG. 13, the surface profile of the glass substrate depicts both depressions and protrusions on the surface of the glass substrate. Additionally, some of the depressions were quite large in nature, up to 2 to 5 nm in depth, with most depressions showing a length of a few micrometers and a depth of a few nanometers.

Example 8—Surface Defects and Cooling Rate

(121) FIG. 14 shows the Zygo reading for a glass substrate that was rapidly cooled upon removal from the bath. As previously discussed, rapid cooling may reduce or restrict the formation of phosphate crystals, and thus, surface defects, on the ion-exchangeable substrate. In Example 8, a glass article was ion exchanged in a 3 kg bath comprising 49% NaNO_3 and 51% KNO_3 . The bath was maintained at a temperature of 380° C. 10 wt % TSP was added to the bath to prevent and remedy lithium poisoning. The glass article was removed from the bath and immediately cooled from a temperature of 380° C. to a temperature of less than 100° C. in one minute. As shown in FIG. 14, the surface defects are much smaller in length and width with very few protrusions.

Example 9—Surface Defects and Bath Temperature

(122) FIG. 15 depicts an in-situ high temperature x-ray diffraction (XRD) pattern as a function of temperature and intensity for a molten salt bath containing KNO_3 , NaNO_3 , and a small amount of lithium sodium phosphate. As shown in FIG. 15, at higher bath temperatures, such as 380° C., the only crystals that were present in the molten salt bath were lithium sodium phosphate, Li_2NaPO_4 salts, which remained in a crystalline form. At lower temperatures, the crystals from the other salts (such as KNO_3 , NaNO_3 and Li_3PO_4) were more prevalent. As the molten salt bath was heated to a temperature of 380° C. and then cooled to room temperature, above 300° C. only lithium sodium phosphate remained present in a crystalline state. Below 300° C., both when the bath was being cooled from 380° C. and when the bath was being heated from room temperature, other crystalline phases existed from crystalline salts present in the bath.

Example 10—Phosphate Salt Reaction Time

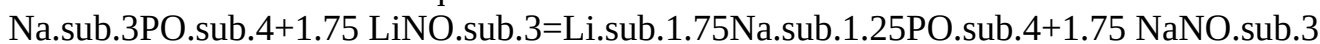
(123) FIG. 16 depicts the differences between adding phosphate salt to a molten salt bath in a powdered, pellet, and capsule formulation, as discussed previously. In Example 10, trisodium phosphate was added to molten salt baths in a powder form, a pellet form, and a capsule form for a treatment time of 75 hours. To produce the capsules, 1 kg of trisodium phosphate, 2 kg KNO_3 , and 1.125 kg of NaNO_3 were mixed thoroughly in a stainless steel container. The mixture was placed in an oven and heated to a temperature of 390° C. The temperature of 390° C. was substantially maintained for 3 hours. The mixture was then removed and cooled to room temperature. Once cooled, the capsule was separated from the stainless steel container.

(124) As shown in FIG. 16, the encapsulated phosphate powder was able to react in the molten salt bath with the lithium nitrate to precipitate out lithium phosphate salts at substantially the same rate as the powdered form. Contrastingly, the pellet form was not able to react as quickly, resulting in a higher concentration of lithium nitrate, present and unreacted in the bath. The capsule did not cloud

the salt bath while still quickly and efficiently reacting with the lithium nitrate, making the capsule a superior choice to the powdered or pellet forms of trisodium phosphate, or other encapsulated phosphate salts.

Example 11—Bath pH

(125) FIG. 17 is a graph of the pH of the salt bath at varying amounts of trisodium phosphate. In Example 11, a 3 kg bath comprising 49% NaNO.sub.3 and 51% KNO.sub.3 was heated and maintained at a temperature of 380° C. Trisodium phosphate (TSP) was added to the bath in accordance with the below equation:



EQUATION 3

(126) The pH was measured by dissolving 5 wt % of the salt in an aqueous solution and measuring the pH at room temperature (approximately 20 to 25° C.). In FIG. 17 the salt bath had a pH of from 6 to 10 as TSP is added to the bath. Without the addition of TSP, the bath had a pH of about 7.8. Upon the addition of TSP, the pH of the bath initially dropped to about 6.3. As more TSP was added to the salt bath, the pH rose to 7.6, then to 9.2, and then 9.8 as the concentration increased, as shown in FIG. 17.

(127) It should be apparent to those skilled in the art that various modifications and variations can be made to the described embodiments without departing from the spirit and scope of the claimed subject matter. Thus, it is intended that the specification cover the modifications and variations of the various described embodiments provided such modification and variations come within the scope of the appended claims and their equivalents.

Example 12—Surface Defects Resulting from Trisodium Phosphate

(128) To determine the effect of trisodium phosphate in the salt bath compositions and, as a result, excess phosphate ions in the molten salt bath, glass-ceramic articles were subjected to two different ion-exchange processes. FIGS. 18A and 18B are optical and SEM images, respectively, of the surface of a 0.6 mm thick glass-ceramic subjected that was subjected to an ion-exchange process at 500° C. for 6 hours with a salt bath composition that did not include trisodium phosphate. As shown by FIGS. 18A and 18B, the surface of the strengthened glass-ceramic article does not have any visible defects. In contrast, FIGS. 18C and 18D are optical and SEM images, respectively, of the surface of a 0.6 mm thick glass-ceramic substrate that was subjected to an ion-exchange process at 500° C. for 6 hours with a salt bath composition that included 0.35 wt % trisodium phosphate. As shown by FIGS. 18C and 18D, the presence of trisodium phosphate in the salt bath composition resulted in surface defects, such as dents and depressions, on the surface of the glass-ceramic article. Moreover, as shown by FIG. 18D, the presence of trisodium phosphate in the salt bath composition resulted in a haze and/or stain on the surface of the glass-ceramic article.

Example 13—Calcium Nitrate Bath Regeneration

(129) To determine the effectiveness of calcium nitrate in precipitating excess phosphate ions in the molten salt bath and, as a result, prevent defects on the surface of ion-exchangeable articles, glass-ceramic articles were subjected to two different ion-exchange processes. FIG. 19A is an SEM image of the surface of a glass-ceramic article that was subjected to an ion-exchange process at 500° C. for 6 hours with a salt bath composition that included 0.12 wt. % trisodium phosphate and 0.09 wt. % calcium nitrate. As shown by FIG. 19A, the surface of the strengthened glass-ceramic article does not have any visible defects. That is, the calcium nitrate effectively reduced the concentration of the phosphate ions in the molten salt bath. This reduction in the concentration of the phosphate ions in turn reduced the formation of tripotassium phosphate crystals and the formation of defects on the surface of the glass-ceramic article. In contrast, FIG. 19B is an SEM image of the surface of a glass-ceramic article that was subjected to an ion-exchange process at 500° C. for 6 hours with a salt bath composition that included 0.12 wt. % trisodium phosphate without any calcium nitrate. As shown by FIG. 19B, defects characteristic of excess phosphate ions in the molten salt bath are present on the surface of the glass-ceramic article.

Example 14—Bath Regeneration with Large-Particle Trisodium Phosphate

(130) A salt bath composition that included 60 wt. % potassium nitrate and 40 wt. % sodium nitrate was prepared. Then, 0.12 wt. % lithium nitrate and 0.5 wt. % silicic acid was added, and the salt bath composition was heated to an ion exchange temperature of 500° C. A batch of 0.6 mm thick glass-ceramic substrates were strengthened (ion-exchanged) in the bath for 6 hours. The strengthened substrates were cleaned and the compressive stress (CS) was measured by ISM and FSM, and the central tension (CT) and the depth of layer (DOL) were measured by FSM (see Table 4, Run 1). Next, the bath was regenerated by adding 0.12 wt. % trisodium phosphate having an average particle size greater than or equal to 100 μm to the bath. After allowing the bath to regenerate for 11 hours, a second batch of 0.6 mm thick glass-ceramic substrates were strengthened (ion-exchanged) in the bath for 6 hours. The strengthened substrates were cleaned and the compressive stress (CS) was measured by ISM and FSM, and the central tension (CT) and the depth of layer (DOL) were measured by FSM (see Table 4, Run 2).

(131) TABLE-US-00004 TABLE 4 ISM FSM CS (MPa) CT (MPa) DOL (μm) CS (MPa) Run 1
219 107 125 206 Run 2 208 106 133 199

(132) The concentration of sodium nitrate, potassium nitrate, and lithium nitrate present in the bath was also measured immediately before and after each run.

(133) TABLE-US-00005 TABLE 5 LiNO₃ (wt. %) NaNO₃ (wt. %) KNO₃ (wt. %)
Before Run 1 0.126 39.4 60.5 After Run 1 0.231 39.4 60.4 Before Run 2 0.166 39.3 60.6 After Run 2 0.273 39.5 60.2

(134) Results showed that lithium cations can be effectively precipitated by the addition of large-particle phosphate salts (in this case, trisodium phosphate). The data indicates that lithium cation concentrations can be significantly reduced within 11 hours at 500° C. Using these baths for ion exchange, the strength of the glass-ceramic substrates processed was maintained over the course of multiple runs despite the release of additional lithium ions into the bath by the glass-ceramic substrates and the presence of phosphate salts within the bath during ion exchange.

Claims

1. A method for regenerating a salt bath composition, the method comprising: heating the salt bath composition to an ion exchange temperature to form a molten salt bath; contacting at least a portion of an ion-exchangeable article comprising lithium oxide with the molten salt bath, whereby lithium cations diffuse from the ion-exchangeable article and into the molten salt bath; adding a first phosphate salt to the molten salt bath, whereby a lithium phosphate salt comprising at least a portion of the lithium cations is formed and precipitates from the molten salt bath; and adding a multivalent salt comprising a multivalent metal cation to the molten salt bath, whereby a second phosphate salt comprising the multivalent metal cation is formed and precipitates from the molten salt bath.
2. The method of claim 1, wherein the salt bath composition comprises: from 40 wt. % to 95 wt. % potassium nitrate based on the total weight of the salt bath composition; from 5 wt. % to 60 wt. % sodium nitrate based on the total weight of the salt bath composition; and from 0.1 wt. % to 1 wt. % lithium nitrate based on the total weight of the salt bath composition.
3. The method of claim 1, wherein the salt bath composition comprises from 0.1 wt. % to 1 wt. % silicic acid based on the total weight of the salt bath composition.
4. The method of claim 1, wherein the ion exchange temperature is greater than or equal to 450° C.
5. The method of claim 1, wherein the ion-exchangeable article comprises from 2 mol. % to 25 mol. % lithium oxide.
6. The method of claim 1, wherein the ion-exchangeable substrate comprises glass or glass-ceramic.
7. The method of claim 1, wherein the first phosphate salt is selected from the group consisting of

trisodium phosphate, tripotassium phosphate, disodium phosphate, dipotassium phosphate, sodium triphosphate, potassium triphosphate, disodium diphosphate, tetrasodium pyrophosphate, potassium pyrophosphate, sodium trimetaphosphate, potassium trimetaphosphate, and combinations thereof.

8. The method of claim 1, wherein a concentration of the lithium cations in the molten salt bath is greater than or equal to a concentration of the first phosphate salt.

9. The method of claim 1, wherein the precipitation of the lithium phosphate salt reduces a concentration of the lithium cations in the molten salt bath by greater than or equal to 0.03 wt. % based on the total weight of the molten salt bath.

10. The method of claim 1, wherein the lithium phosphate salt comprises at least one of insoluble trilithium phosphate, insoluble dilithium sodium phosphate, or insoluble lithium disodium phosphate.

11. The method of claim 1, wherein the multivalent metal cation is selected from the group consisting of magnesium, calcium, aluminum, strontium, barium, iron, and combinations thereof.

12. The method of claim 1, wherein the multivalent salt comprises calcium nitrate.

13. The method of claim 1, wherein the precipitation of the second phosphate salt reduces a total concentration of phosphate salts in the molten salt bath to less than 0.01 wt. % based on the total weight of the molten salt bath.

14. The method of claim 1, further comprising removing the ion-exchangeable article from contact with the molten salt bath, wherein the ion-exchangeable article is removed before adding the first phosphate salt and adding the multivalent salt.

15. The method of claim 1, wherein the multivalent salt is added to the molten salt bath in an amount from 0.01 wt. % to 0.5 wt. %.
